Department of Electronics & Electrical Communication Engineering
Indian Institute of Technology Kharagpur

Multimedia Systems And Applications (EC60104)

Assignment 3 : K-L Transformations

Group No. : 1
Members: Adipta Halder (21EC37023)
Mayukh Shubhra Saha (21EC37009)
Abhradeep De (21EC37022)

PROBLEM STATEMENT

(a) Consider the first six frames of the video sequence “Foreman”. Compose 6-element vectors by picking up pixel values at the same spatial position over six consecutive frames.

(b) Determine the mean of these 6-element vectors, considering all spatial positions. Compute the 6x6 covariance matrix and determine its eigenvalues and the corresponding eigenvectors. Retain only top two of these eigenvalues and the corresponding eigenvectors

(c) Obtain the top two principal component images by projecting the vectors (obtaining dot-products) on the two principal eigenvectors and display the results.

(d) Apply inverse K-L transformation and obtain the reconstructed frames. Compute the PSNR of each reconstructed frame.

BRIEF THEORY

Transform coding is a source coding technique that exploits statistical dependencies in signals by mapping the original data into a new domain where the components are decorrelated and energy is concentrated in a small number of coefficients. Given an input vector x, a linear transform (T) is applied to obtain the coefficient vector (y = Tx). If the transform is properly chosen, the signal energy becomes concentrated in a few coefficients while the remaining coefficients have small magnitudes. This enables efficient compression: high-energy coefficients are preserved with higher precision, while low-energy coefficients can be coarsely quantized or discarded. After quantization and entropy coding, the inverse transform reconstructs an approximation of the original signal.

The Karhunen–Loève (K–L) transform is the optimal linear transform for decorrelating a random vector in the mean-square error sense. For a zero-mean random vector x with covariance matrix Rx, the K–L transform matrix is formed by the eigenvectors of Rx. If $\mathbf{Rx = ULU^T}$, where U contains the orthonormal eigenvectors and L is a diagonal matrix of eigenvalues, then the transform $\mathbf{y = U^T x}$ produces uncorrelated coefficients whose variances equal the eigenvalues. Because the eigenvalues typically decay rapidly, most of the signal energy is concentrated in the first few principal components, enabling dimensionality reduction and efficient coding. Retaining only the **largest k eigenvalues** and their eigenvectors yields the minimum mean-square reconstruction error among all k-dimensional linear approximations, which makes the K–L transform theoretically optimal for transform coding of correlated data.

SOLUTION

NOTE : All relevant scripts, input and output files are present in the GitHub repository, under the directory **"Assignment_3"**. Inside this folder, there is also a detailed **README.md** file which explains the file structure, file contents and commands to run the scripts.

GitHub Repo Link : https://github.com/mayukhss262/Multimedia_Assignments

Code Explanation

All four sections of the problem are implemented using a single Python script, with multiple functions. The script depends on NumPy for linear algebra operations and Pillow (PIL) for image I/O. It then defines four constants:

- SCRIPT_DIR: the current working directory, used as the base path for all file operations.
- FRAME_DIR: subdirectory **foreman_frames/**, expected to contain the 6 input frames named **foreman_frame_001.png through foreman_frame_006.png**.
- OUTPUT_FILE: path to **results.txt**, where all numerical output (mean vector, covariance matrix, eigenvalues/eigenvectors, PSNR) is written.
- OUTPUT_IMG_DIR: subdirectory **output_images/**, to store generated images.

Each frame is loaded as a float64 2D array (height × width). The spatial dimensions H and W are extracted from the first frame, and **N = H * W** denotes the total number of pixel positions (i.e., the dimensionality of each observation vector). The 6 frames are then flattened into 1D row vectors and stacked vertically to form the **data matrix X** of shape (6, N). Here, each row represents one frame's pixel intensities as a single vector, and each column represents the intensity values of a single spatial location across all 6 frames. This formulation treats the 6 frames as 6 random variables observed at N spatial sample points - the standard setup for the K-L transform. The mean vector **μ** is computed by averaging each row of X across all N spatial positions. This yields a 6-element vector where each entry is the mean pixel intensity of the corresponding frame. Mean-centering is then performed by broadcasting this column vector and subtracting it from every column of X, producing X_centered. This is a prerequisite for covariance estimation and ensures the K-L transform operates on zero-mean data.

The 6×6 covariance matrix C is computed as $(1/N) \cdot \tilde{X} \cdot \tilde{X}^T$, where $\tilde{X}$ is the mean-centered data matrix. Each entry C[i, j] measures the spatial covariance between frames i and j. Division by N (the number of spatial samples) yields the biased (population) covariance estimate. This matrix is symmetric and positive semi-definite by construction - properties that guarantee real, non-negative eigenvalues.

For eigenvalue decomposition, **np.linalg.eigh** is used (instead of np.linalg.eig) because the covariance matrix is guaranteed symmetric - eigh exploits this structure for better numerical stability and always returns real eigenvalues. By default, eigh returns eigenvalues in ascending order, so the script reverses the sort to obtain **descending order** (largest eigenvalue first). The eigenvectors (columns of the eigenvector matrix) are reordered correspondingly. The top 2 eigenvectors are then extracted as the retained basis vectors; these correspond to the directions of maximum variance in the 6-frame space. The remaining 4 components are discarded.

The forward K-L transform projects the centered data onto the retained eigenvectors: $\mathbf{Y} = \mathbf{A}^T \cdot \tilde{\mathbf{X}}$, where A is the (6, 2) matrix of top eigenvectors. The result has shape (2, N) - each row holds one principal component's coefficients across all spatial positions. To produce a visual representation, each principal component vector is min-max normalized to the [0, 255] range and reshaped back to (H, W) for saving as an 8-bit grayscale PNG. The normalization is necessary because raw principal component values span an arbitrary real-valued range and cannot be directly interpreted as pixel intensities.

The inverse K-L transform reconstructs the data: $\hat{\mathbf{X}} = \mathbf{A} \cdot \mathbf{Y} + \boldsymbol{\mu}$. This maps the 2-component representation back into the original 6-frame space and adds back the mean vector. The result is a (6, N) matrix of reconstructed pixel values. Since only 2 of 6 components are used, this is a lossy reconstruction. For each frame, the reconstructed vector is clipped to [0, 255], reshaped to (H, W), cast to uint8, and saved as a PNG. Finally, PSNR of the reconstructed PNG frames are computed.

All results, including the mean and covariance matrices, eigenvalues and corresponding eigenvectors and the PSNRs of each reconstructed frame are written to the output file **results.txt**.

RESULTS

Input Frames

- All frames are in 8-bit monochrome (grayscale) and each frame has a size of **352 * 288** pixels.

First frame of Foreman video sequence (foreman_frame_001.png)

Second frame of Foreman video sequence (foreman_frame_002.png)

Third frame of Foreman video sequence (foreman_frame_003.png)

Fourth frame of Foreman video sequence (foreman_frame_004.png)

Fifth frame of Foreman video sequence (foreman_frame_005.png)

Sixth frame of Foreman video sequence (foreman_frame_006.png)

Top Two Principal Component Images

First principal component image, corresponding to largest eigenvalue (principal_component_1.png)

Second principal component image, corresponding to second largest eigenvalue (principal_component_2.png)

Reconstructed Frames

First reconstructed frame of Foreman video sequence (reconstructed_frame_001.png)

Second reconstructed frame of Foreman video sequence (reconstructed_frame_002.png)

Third reconstructed frame of Foreman video sequence (reconstructed_frame_003.png)

Fourth reconstructed frame of Foreman video sequence (reconstructed_frame_004.png)

Fifth reconstructed frame of Foreman video sequence (reconstructed_frame_005.png)

Sixth reconstructed frame of Foreman video sequence (reconstructed_frame_006.png)

PSNRs of Reconstructed Frames

Frame Number	PSNR (dB)
001	30.8204
002	35.1673
003	30.0922
004	30.9812
005	36.1683
006	31.2628

Observations

The per-frame mean intensities of the input frames are tightly clustered around 160.2–160.8, indicating minimal global brightness variation across the 6 frames. This is expected for consecutive frames in a video sequence with stable illumination. The 6×6 covariance matrix exhibits high off-diagonal values (in the range 2600–2900), with diagonal entries (individual frame variances) only marginally larger than the off-diagonal terms. This confirms **strong inter-frame correlation** - adjacent frames share most of their spatial structure, which is the fundamental property that makes K-L transform effective for temporal decorrelation of video.

The total energy (sum of all eigenvalues) is 17,413.6. The top 2 eigenvalues together account for nearly 98.5% of this total energy. The remaining 4 components collectively hold only around 1.5% of the energy. This extreme concentration implies that the remaining 4 components only hold negligible information and can be discarded without visible loss of image information.

All reconstructed frames achieve PSNR values around 30-36dB. This represents an acceptable quality of reconstruction. It is seen that frames which capture temporal activity, such as unique motion or structural changes, have lower PSNR values - these frames contain unique information not captured by any other adjacent frames.